Assignment No. 11

1-D Diffusion Problem

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FlowThermoLab

Objective

Objective of this study is to

1. Show temperature distribution of 1D SS heat conduction problem
2. Temperature distribution for 1D transient heat conduction using FTCS Scheme

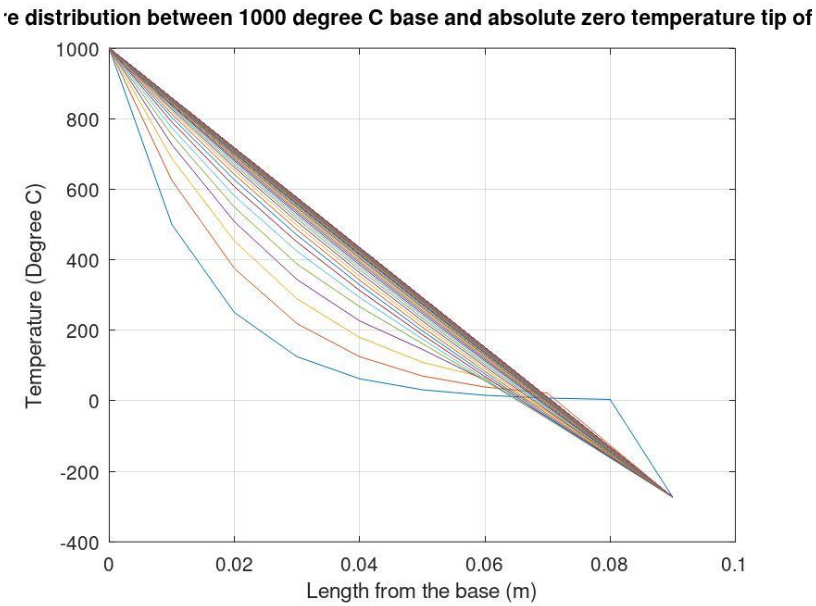
1. Problem Statement

(a). A 10 cm rod with base temperature 1000 degree C and tip at absolute zero temperature

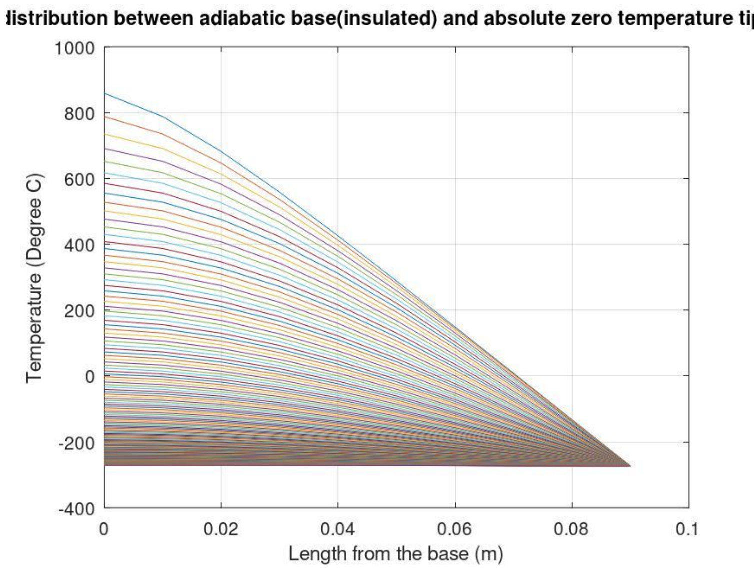
(b). Insulated base with absolute zero at tip

```
clc; clear all; close all;
l = 0:.01:(10/100 - .01); % different lengths from the base(m)
n = 10;    % number of nodes

T = zeros(n,1); %initial guess values

Tb = 1000; % base temperature
Ttip = -273.16;    % tip temperature


% solution

k = 200; %number of iterations

%case 1
for j=1:1:k

  T(1,1) = Tb; %first node

  for i = 2:1:n-1
    T(i,1) = (T(i+1,1) + T(i-1,1)) / 2; % temp inside the domain
  end

  T(n,1) = Ttip; %last node


  figure(1);
  plot(l, T); hold on;
  title("Temperature distribution between 1000 degree C base and absolute zero temperature tip of a straight rod")
  xlabel("Length from the base (m)");
  ylabel("Temperature (Degree C)");
  grid on;

end

hold off;

%case 2

for j=1:1:k

  T(1,1) = T(2,1); %first node

  for i = 2:1:n-1
    T(i,1) = (T(i+1,1) + T(i-1,1)) / 2; % temp inside the domain
  end

  T(n,1) = Ttip; %last node

  figure(2)
  plot(l, T); hold on;
  title("Temperature distribution between adiabatic base(insulated) and absolute zero temperature tip of a straight rod")
  xlabel("Length from the base (m)");
  ylabel("Temperature (Degree C)");
  grid on;

end

hold off;
```

Results



e distribution between 1000 degree C base and absolute zero temperature tip of
Temperature (Degree C)
Length from the base (m)



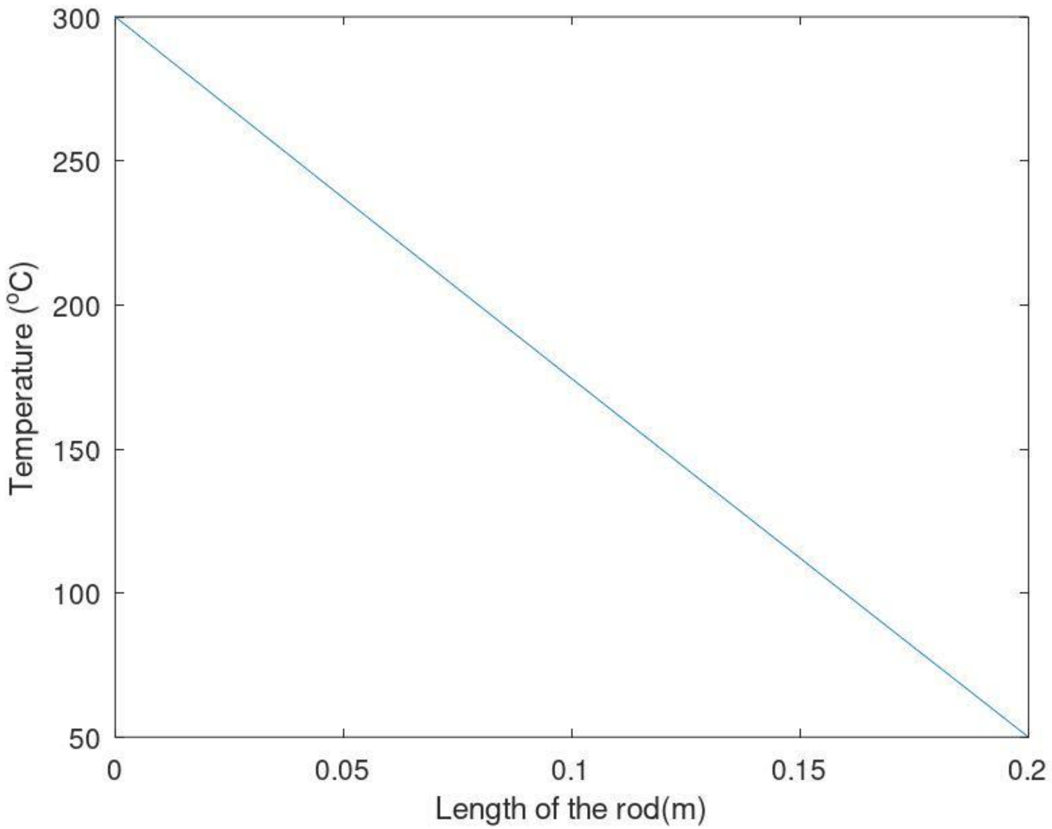
listribution between adiabatic base(insulated) and absolute zero temperature tip
Temperature (Degree C)
Length from the base (m)

2. Problem Statement

A 20 cm rod with base temperature 300 degree C and tip at 50 degree C. Diffusion Co-efficient is 0.05. Initial temperature across the rod: 30 degree C

```
% 1d Heat heat conduction problem using FTCS scheme for transient problem
clc; clear all; close all;


% Geometry

xmin = 0;
xmax = 0.2; % rod length(meter)
N = 20     %number of nodes
dx = (xmax - xmin) / (N - 1) %grid space
x = xmin:dx:xmax;
dt = 1e-3
tmax = 0.25; % total simulation time
t = 0:dt:tmax;
alpha = 0.05; %diffusion coefficient

% Initial and Boundary Conditions

Tcurrent = ones(1,N)* 30; %initial temperature across the rod
Tbase = 300;
Ttip = 50;

% Solution

d = (alpha * dt) / (dx^2)  %for parabolic equation as such

for j=2:length(t) % loop for time step

   T = Tcurrent;

   for i=1:N      % Space stepping

     if i == 1
       T(i) = Tbase;
     elseif i == N
       T(i) = Ttip;
     else
       T(i) = T(i) + (d * (T(i + 1) - 2 * T(i) + T(i - 1)));
     end

   endfor

   Tcurrent = T;
   time = j*dt

   plot(x, Tcurrent)
   xlabel("Length of the rod(m)")
   ylabel("Temperature (^{o}C)")
   shg
   pause(0.01)

end
```

Results



300
250
200
150
100
50
Temperature (°C)
0
0.05
0.1
0.15
0.2
Length of the rod(m)

Discussion

The simulation reaches Steady State at approximately 0.15 seconds. Initially 10 number of nodes were used, that resulted in grid space of 0.022. Taking time step size 10e-3 we got d = $0.1013<0.5$,thus stable.

Then, 20 number of nodes were used, that resulted in grid space of 0.0105. Taking time step size 10e-3 we got d = $0.4513<0.5$,thus still stable.

Later, 30 number of nodes were used, that resulted in grid space of 6.89e-03. Taking time step size 10e-3 we got d = $1.0512>0.5$,thus unstable and the simulation failed. To stabilize it time step was reduced to 0.1e-3, resulting in d being $0.1051<0.5$, thus stable. But reducing the time step size elongated the simulation time